

FitBuddy AI — Solution Architecture

Phase 3 • Project Design Phase

Architecture layers

Presentation layer

- index.html for the main user page; coach.html for the coach dashboard; styles.css for responsive styling; app.js and coach.js for browser-side API interaction.

Application/API layer

- app/main.py provides the FastAPI application entry point and route definitions.

Configuration and validation

- app/config.py manages application/environment configuration; app/schemas.py defines structured API request/response schemas.

Persistence

- app/database.py handles SQLite setup/connection; app/models.py contains database model definitions.

Business logic

- app/services.py contains core service/business logic, generation, and feedback-based adjustments.

Testing

- tests/test_api.py contains automated API/application tests.

Main modules

Module	Responsibility
app/main.py	FastAPI entry point and routes
app/config.py	Environment/application configuration
app/database.py	SQLite setup and connections
app/models.py	Database models
app/schemas.py	API schemas
app/services.py	Generation and feedback adjustment logic
app/static/*	User and coach frontend
tests/test_api.py	Automated tests